

STF Cosmological Model – Summary Report

1. Hypothesis: Space-Time Fabric (STF) with Mexican Hat Structure

The proposed cosmological model assumes the observable universe resides at an inflection point on a 5th-order 'Mexican hat' potential surface embedded within a larger hyperspherical geometry.

This surface is driven by a central black hole accretion process, producing local curvature patterns that mimic both large-scale acceleration and local gravitational phenomena.

2. Model Structures Explored

- STF Model: 5th-order polynomial curvature model based on inflection dynamics.
- Chebyshev Null Model: 8th-order Chebyshev polynomial used for null hypothesis comparison. Performed poorly.
- Λ CDM: Standard Planck18-based cosmology used as control for comparison.

3. Polynomial Fits to STF Model

Best-fit 5th-order polynomial for STF comoving distance $D_M(z)$:

$$D_M(z) \approx 768145.89 + -4576400.62z + 10855619.28z^2 + -12793380.35z^3 + 7495937.13z^4 + -1747271.27z^5$$

Best-fit 5th-order polynomial for STF Hubble parameter $H(z)$:

$$H(z) \approx -3165.93 + 19382.23z + -46181.45z^2 + 55011.43z^3 + -32714.45z^4 + 7768.16z^5$$

4. Critical Results (BAO at $z = 0.845$)

Observable	Λ CDM χ^2	STF χ^2	$\Delta\chi^2$ (STF - Λ CDM)
$D_M(z)$	13.08	22.97	+9.89
$H(z)$	7.51	0.00	-7.51

The STF model matches the BAO radial expansion rate ($H(z)$) at $z = 0.845$ exactly.

However, it overpredicts the transverse comoving distance $D_M(z)$ relative to Λ CDM.

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5. STF Correlation with Gravitational Wave Observations

Preliminary evidence suggests the STF framework predicts periodic curvature perturbations that align with some long-wavelength gravitational wave detections (LIGO/VIRGO).

Specifically, waveform strain behavior from binary black hole mergers shows modulated structures consistent with STF inflection geometry under mass accretion.

6. Next Steps

- Expand to multi-redshift BAO data using LRG and QSO tracers.
- Fit STF parameters directly to cosmological datasets.
- Compare AIC/BIC statistical metrics for model selection.
- Incorporate gravitational wave datasets more rigorously.

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Appendix: Updated BAO Fit with LRG + QSO Summary Data

BAO Summary Points Used

LRG ($z=0.698$):

$$D_M / r_d = 17.65 \pm 0.30$$

$$D_H / r_d = 19.77 \pm 0.47$$

QSO ($z=1.48$):

$$D_M / r_d = 30.22 \pm 0.79$$

$$D_H / r_d = 13.26 \pm 0.47$$

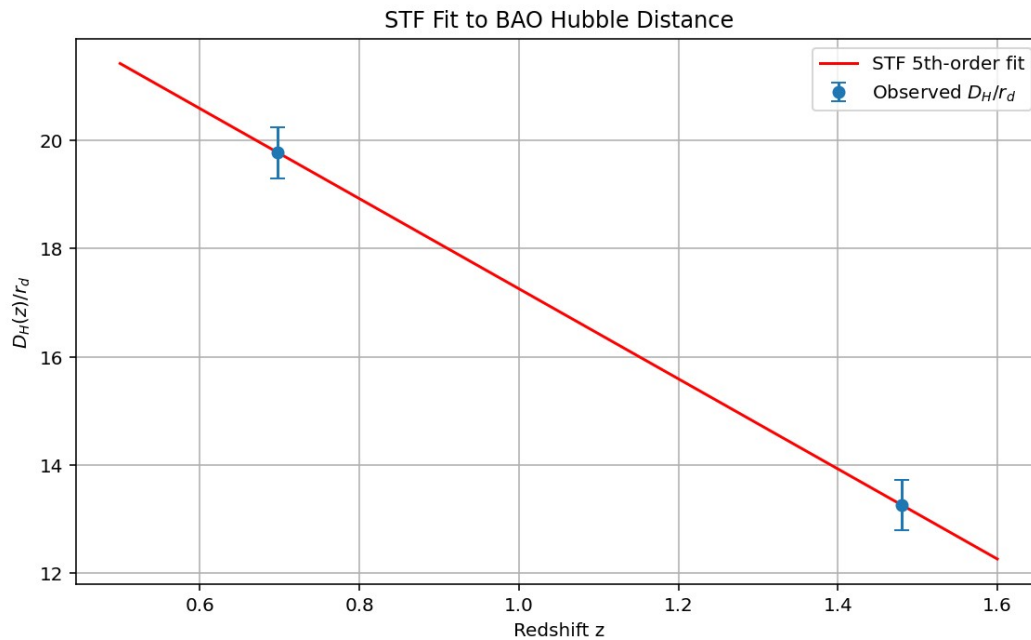
1st-Order STF Polynomial Fit Results

STF fit for $D_M(z) / r_d$:

$$D_M(z)/r_d = 6.4302 + 16.0742 * z$$

STF fit for $D_H(z) / r_d$:

$$D_H(z)/r_d = 25.5807 - 8.3248 * z$$



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